

IT603 - Introduction to programming

Course Placement: Autumn 2025 – Core MSc IT

Course Format: It is 3 lectures and one lab every week

Prerequisites (if any): None

Credit Structure (L-T-P-Cr): 3-0-2-4

Instructor's Name with email: Ankush Chander (ankush_chander@dau.ac.in)

Course Objective: Course is designed as a first timer introduction to programming. It assumes no prior exposure to programming. Course aims at imparting students solid understanding of not only programming language features but also fundamentals knowledge that" transferable to other languages also.

Course Content:

Introduction to Linux environment - Computer system, Shell , Command line Interface, Standard I/O, System Path
Introduction to Programming – Introduction to Modern C++, concept of program flow, why modern C++ without OOP is C, Hello world, Program structure, Compilation process
Data types and Concept of Variables – Data types, Variable, Static variables, global variables
Program Control – Logical operators, bitwise operators, functions, Conditional constructs
Array, Pointers, Structures – Arrays, pointers, array of characters, multi-dimensional arrays, pointers to functions, structures and classes
Dynamic Memory Allocation – creating variables on the free-store, memory management
Modern C++: Templates – Generic programming, Template structure and definition
Modern C++: STL – std::string, STL algorithms

Suggested Textbook/references:

1. [Think C++](#) by Allen B Downey
2. [the C++ Programming Language](#) by Bjourne Stroustrup
3. UNIX : Concepts and Applications, Sumitabha Das
4. Computer Systems: A Programmer's Perspective, Randal E. Bryant and David R. O'Hallaron

Assessment Method:

- Labs and Assignments: 30
- In-Sem Exam: 30
- End Sem Exam: 40

Grading Policy: Relative grading with the following slabs:

- >90 = AA
- 80-90 = AB
- 65-79 = BB
- 50-64 = BC
- 40-50 = CC
- <39 = CD

Course outcome:

After taking this course, students will be able to write portable, efficient application programs in C++. They will also be able to navigate Linux environments fluently and take up new programming languages with relative ease.